

Advanced Econometrics Lecture 09

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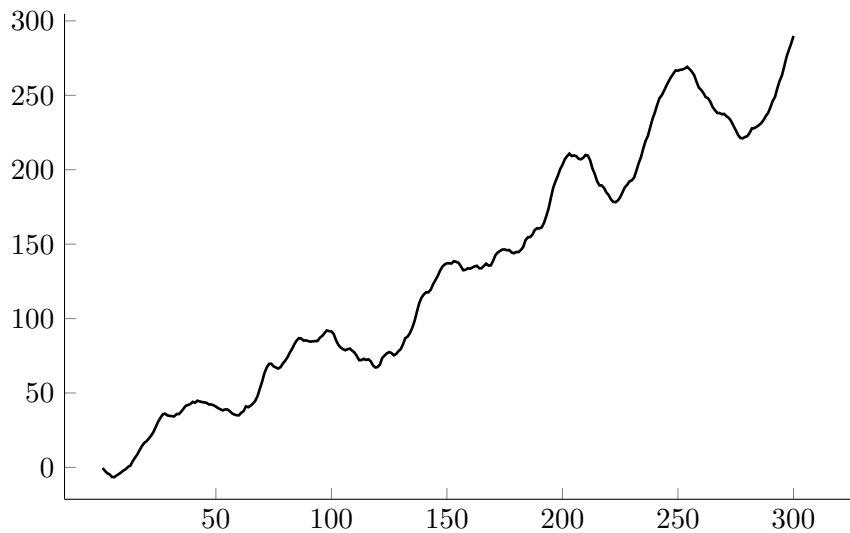
Faculty of Economic Sciences, University of Warsaw

Warsaw, 28-04-2020

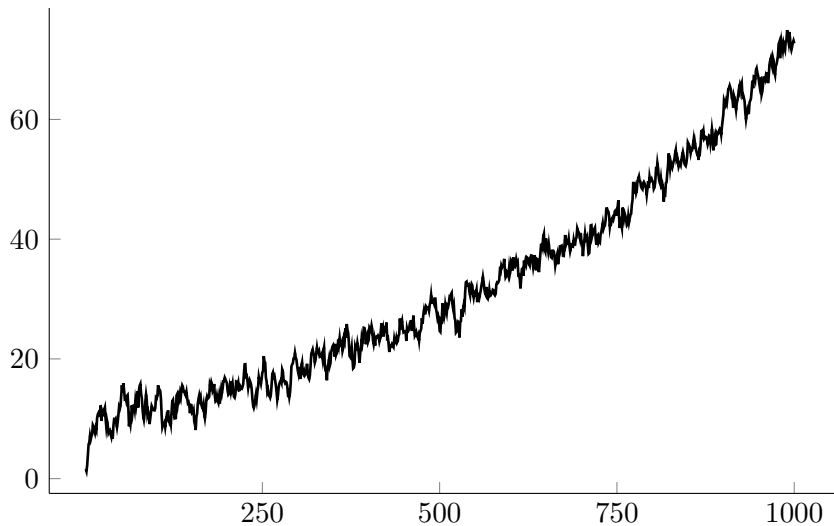
Stationarity

- a) Intuitively, a time series is stationary if its properties does not depend on time.
- b) The process for y_t is *weakly stationary* if neither the mean μ_t nor the autocovariances $Cov(y_t, y_{t+h})$ depend on the date t :
 - the expected value of y_t is finite and constant in time $\mathbb{E}(y_t) = \mu < \infty$
 - its variance y_t is finite and constant in time $Var(y_t) = \sigma^2 < \infty$
 - covariance depends only on t that $Cov(y_t, y_{t+h}) = Cov(y_s, y_{s+h}) = \gamma_h$ for any t, s, h .
- c) The assumption of the stationarity is necessary for deriving the distributions of test statistics used in hypotheses testing.

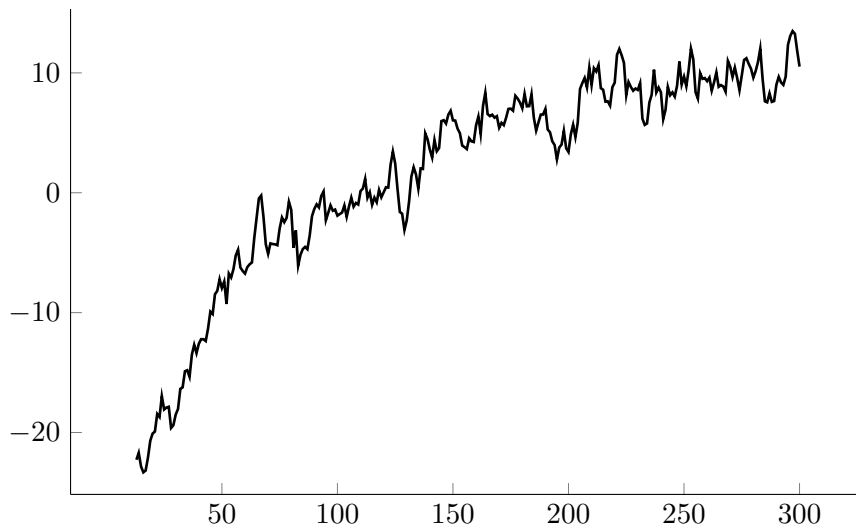
Example 1



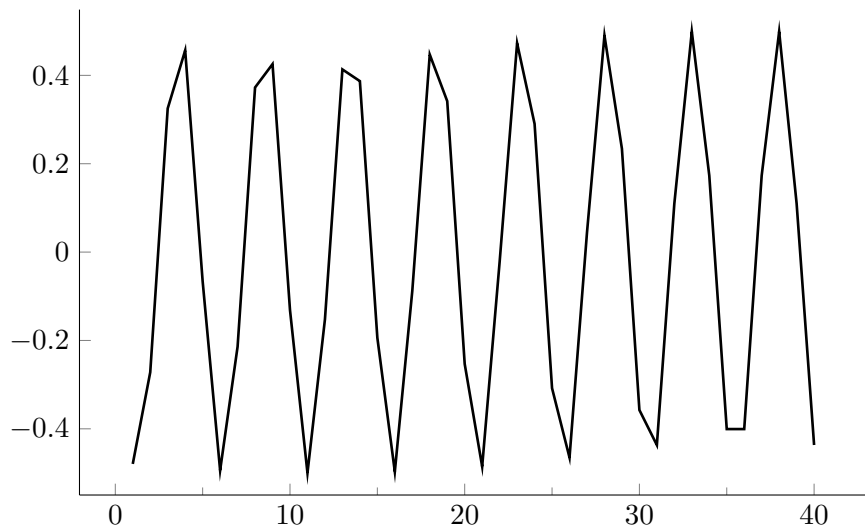
Example 2



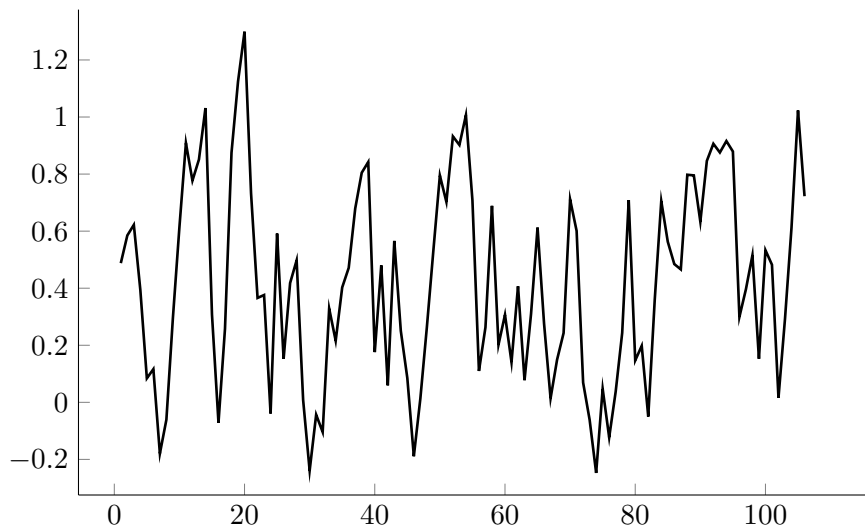
Example 3



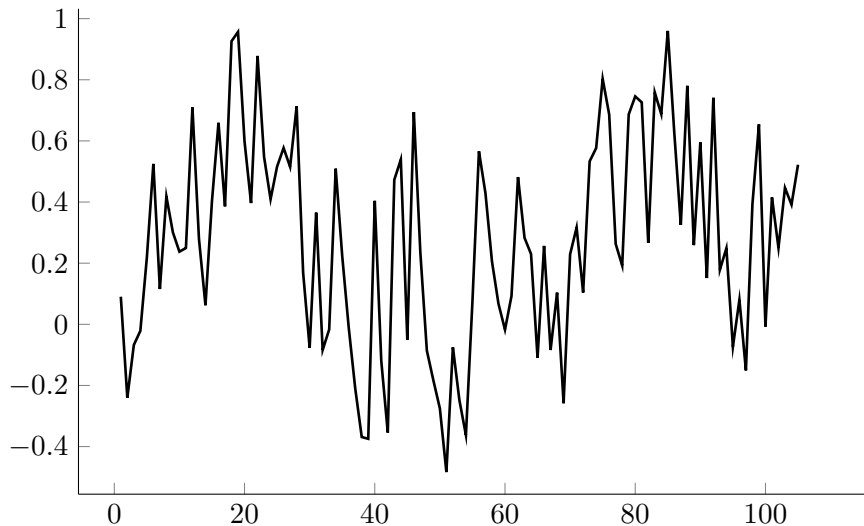
Example 4



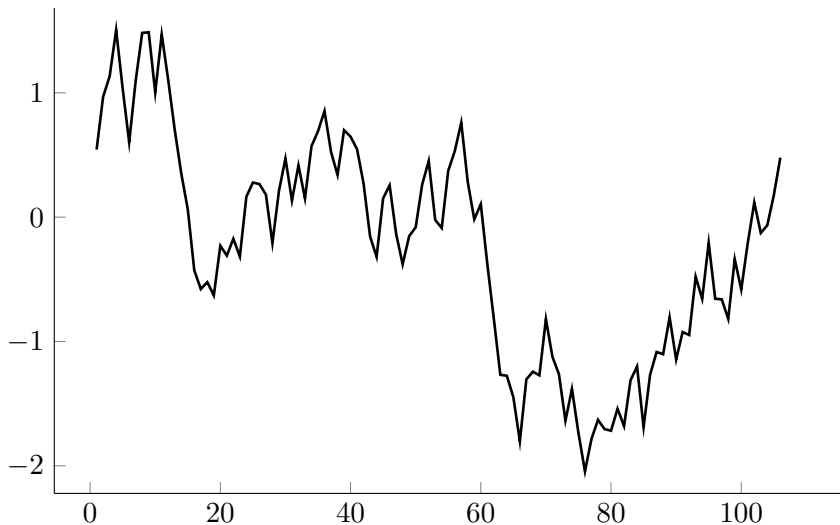
Example 5



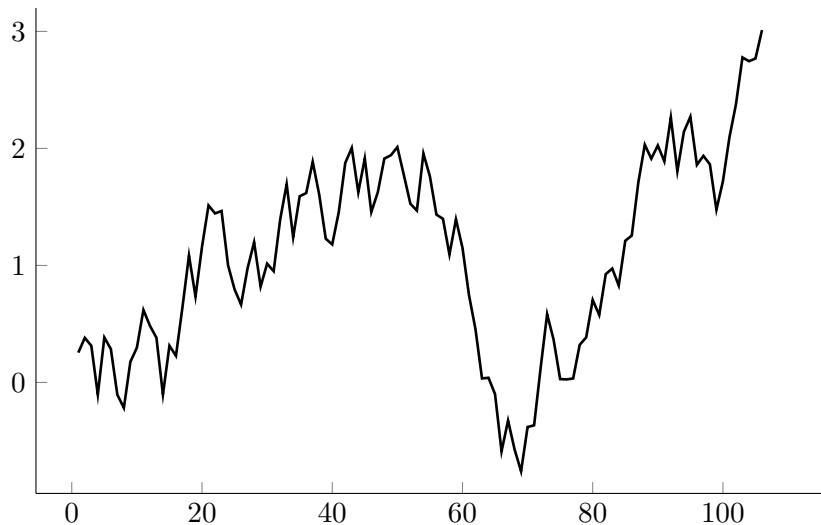
Example 6



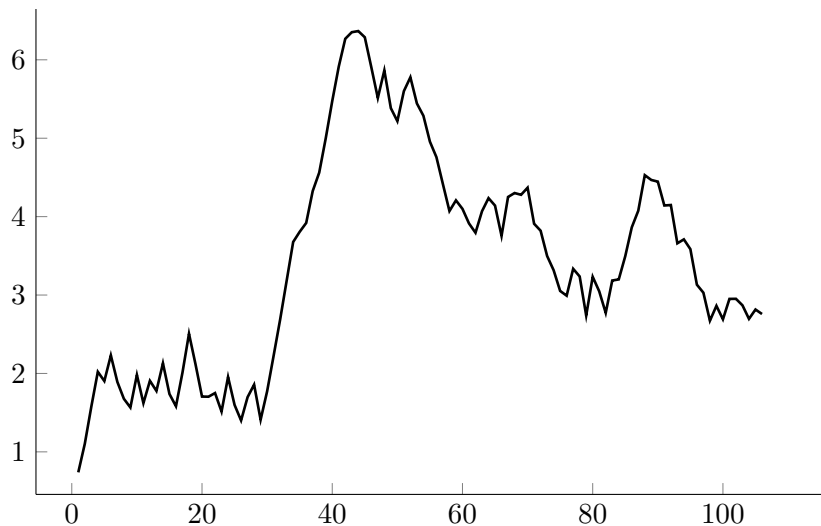
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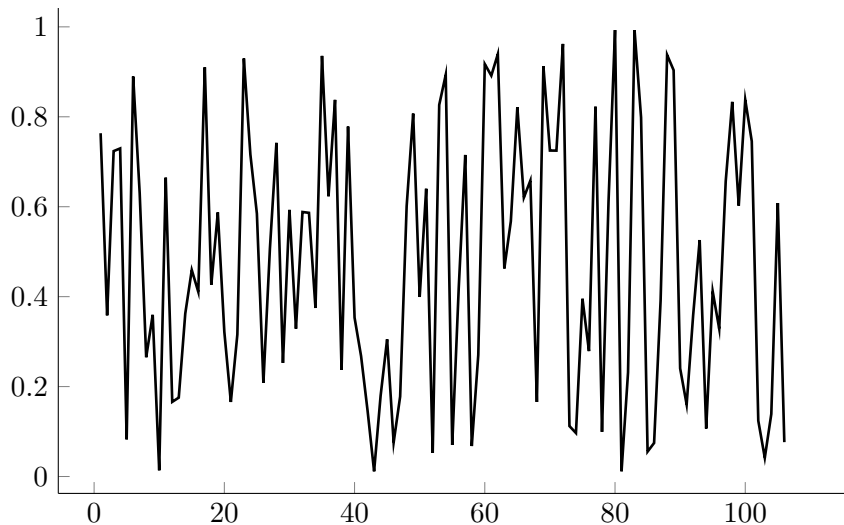
Example 8



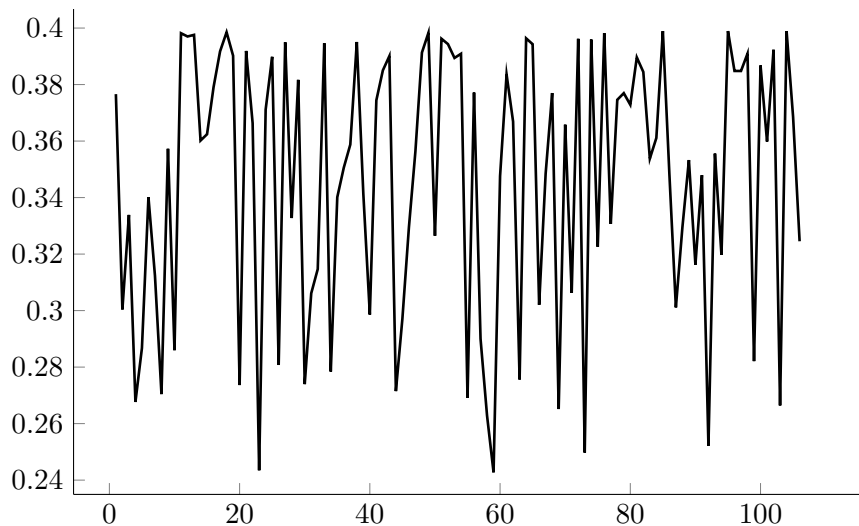
Example 9



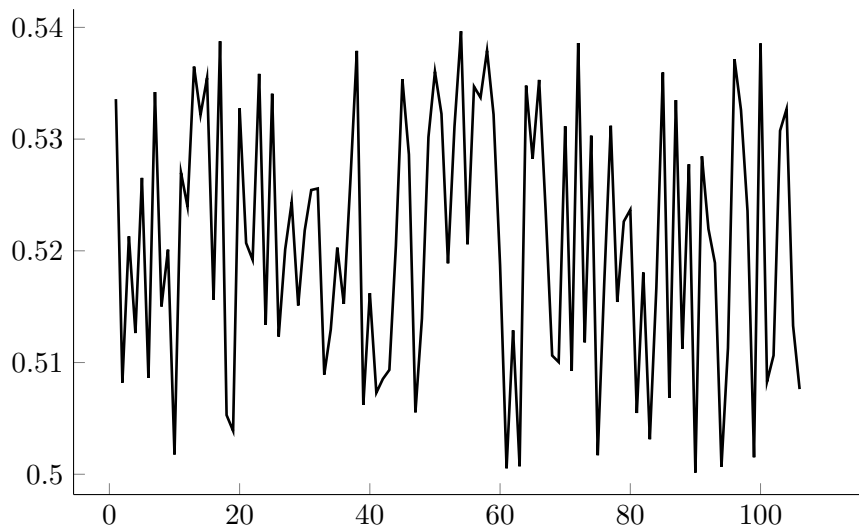
Example 10



Example 11



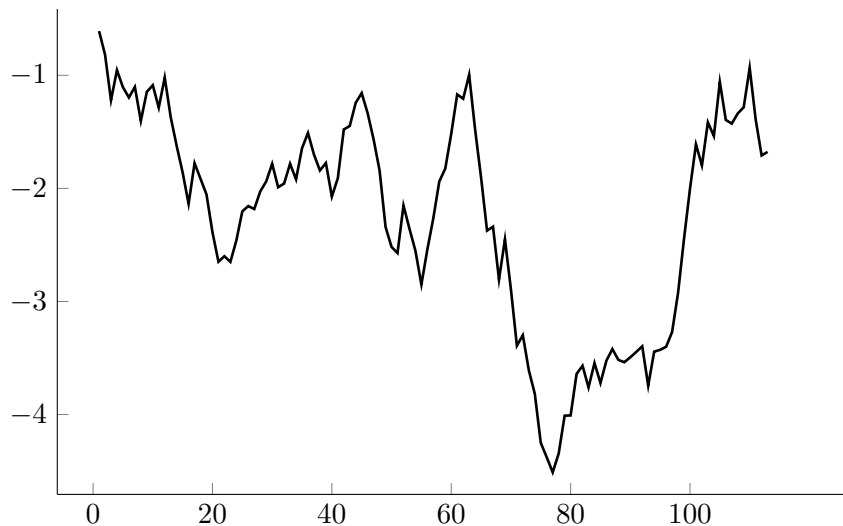
Example 12



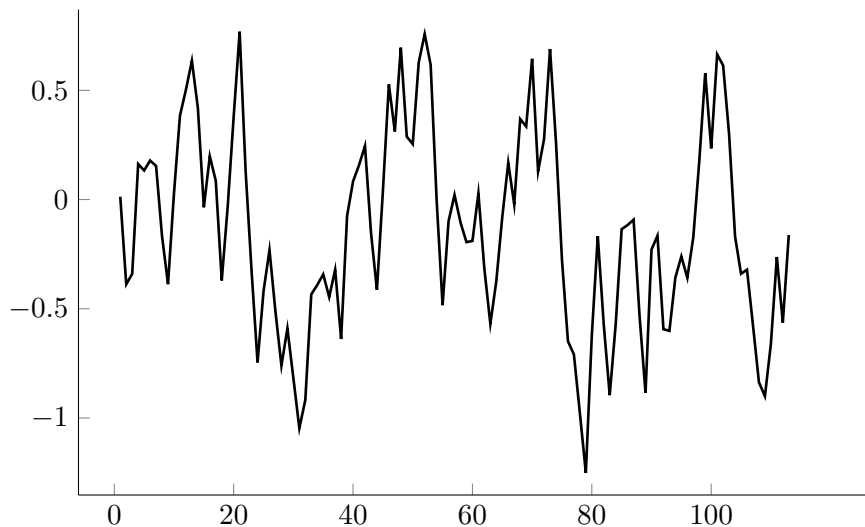
Stationarity

- a) White noise is stationary.
- b) Random walk is non-stationary.
- c) For exercises 1 - 12 assess stationarity.

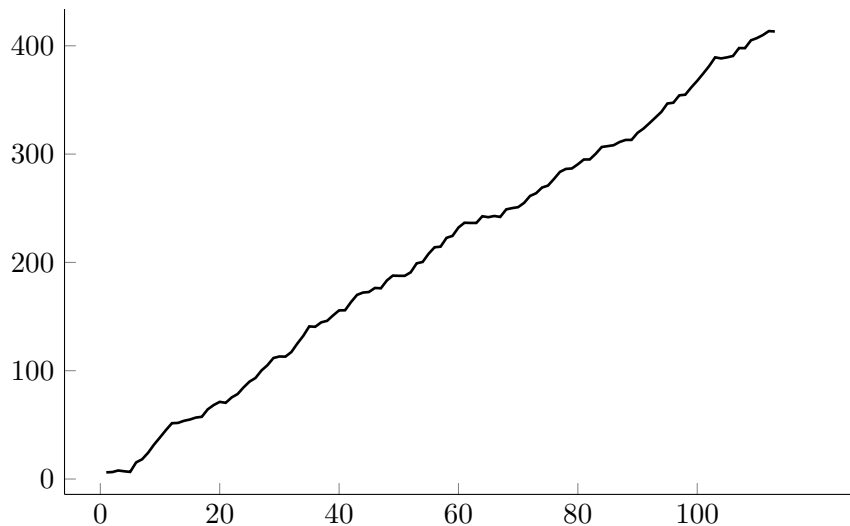
Exercise 1



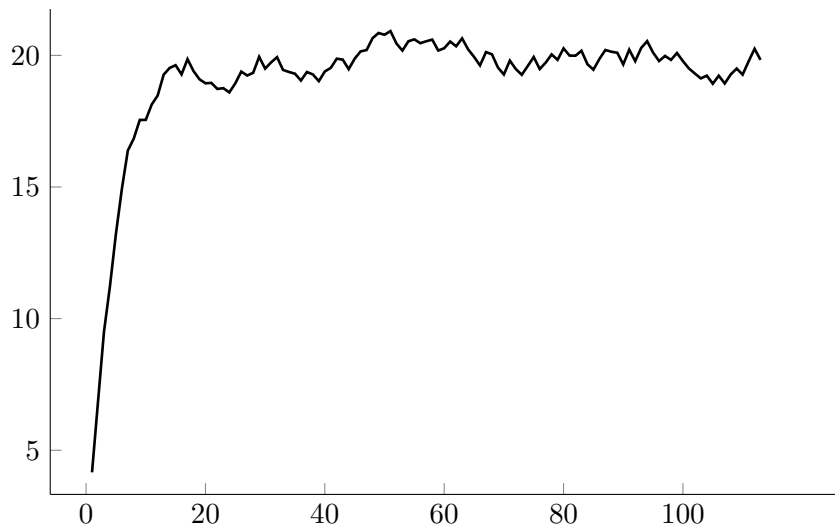
Exercise 2



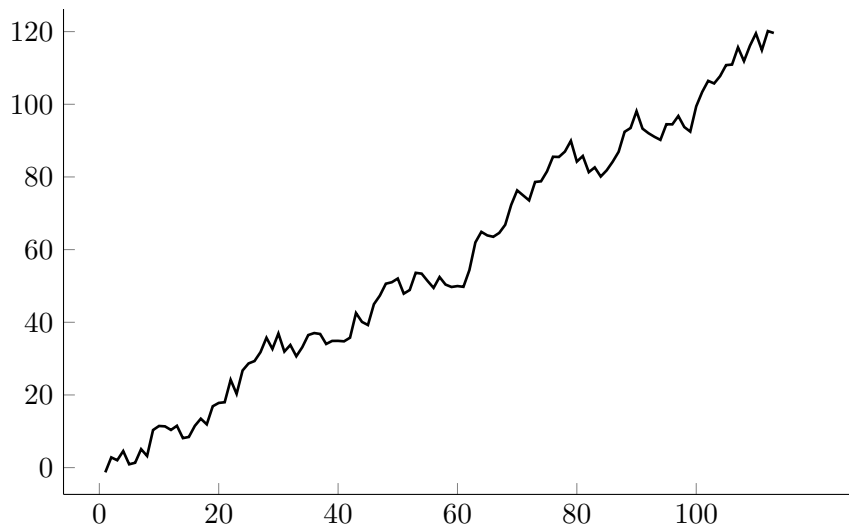
Exercise 3



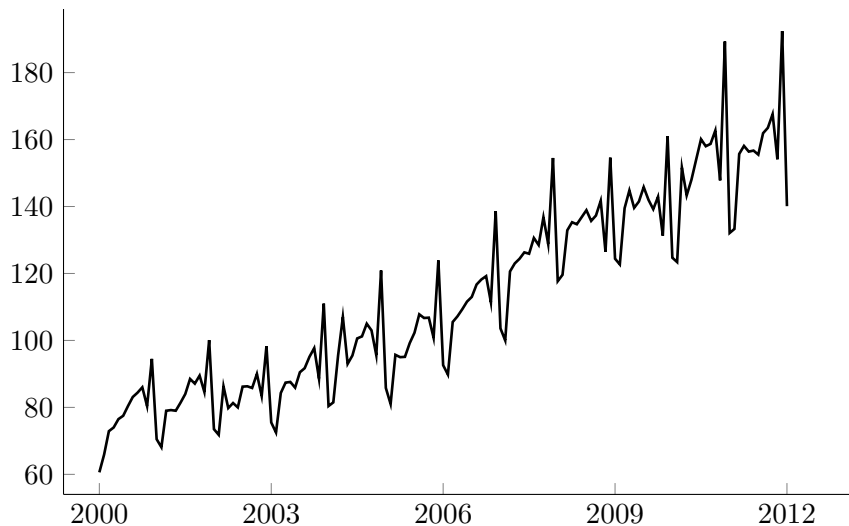
Exercise 4



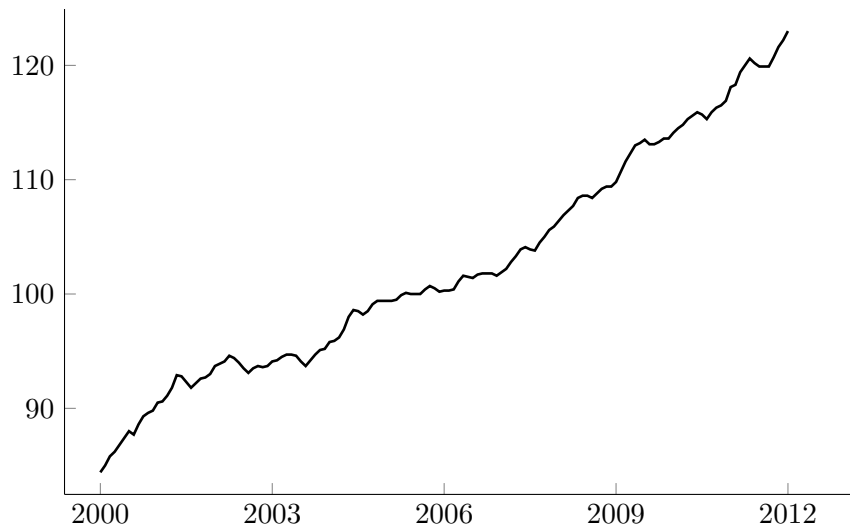
Exercise 5



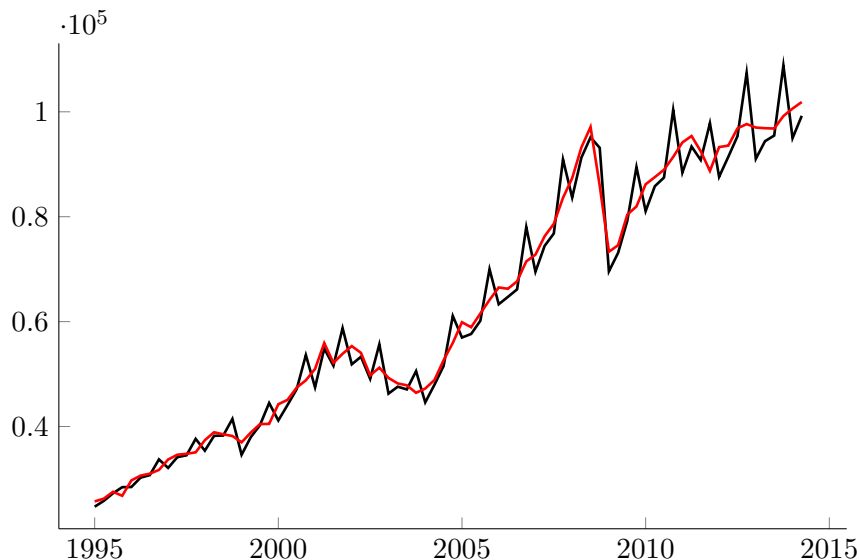
Exercise 6 – Retail trade index (POL)



Exercise 7 - HICP (POL)



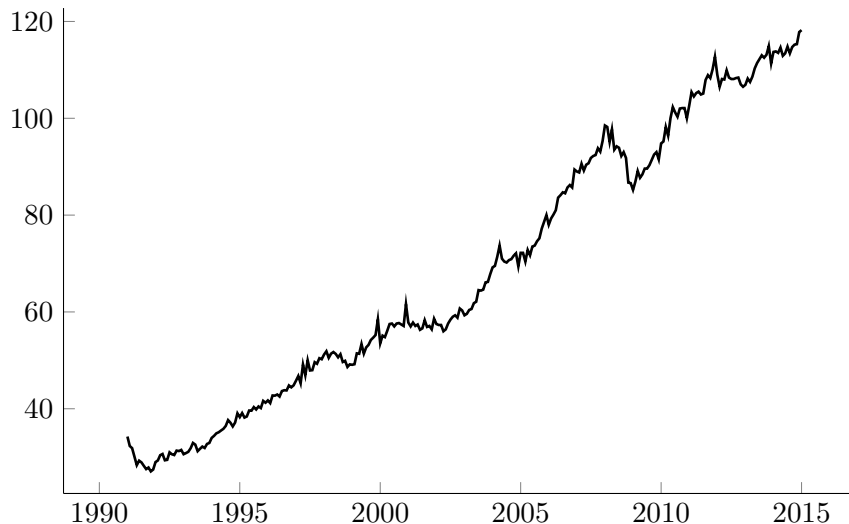
Exercise 8 – GDP (POL) – NSA and SA – ESA 95



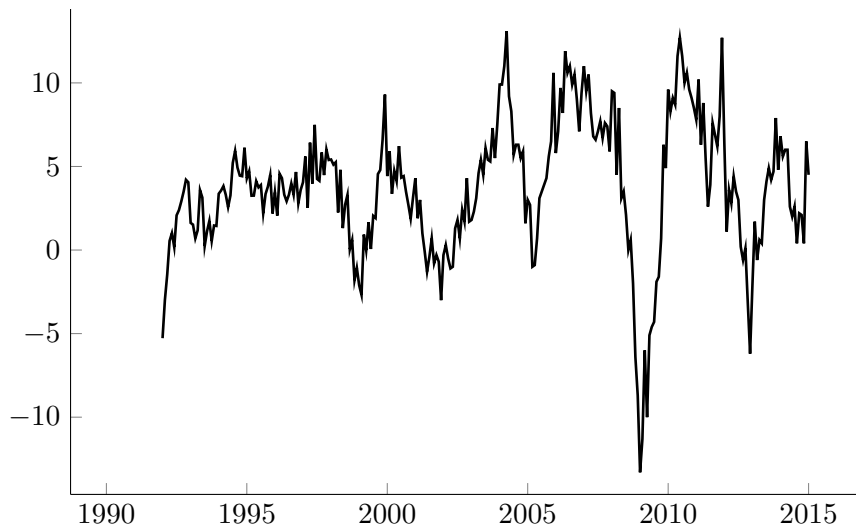
Exercise 9 – Consumer confidence current index (POL)



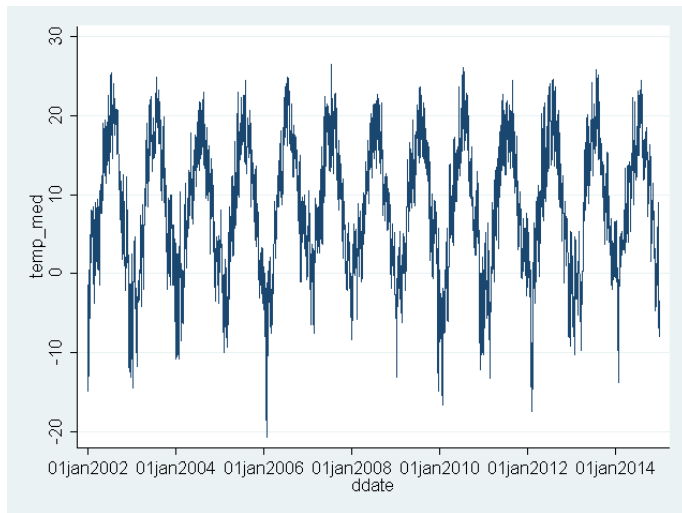
Exercise 10 - Industrial production index



Exercise 11 - Δ_{12} Industrial production index



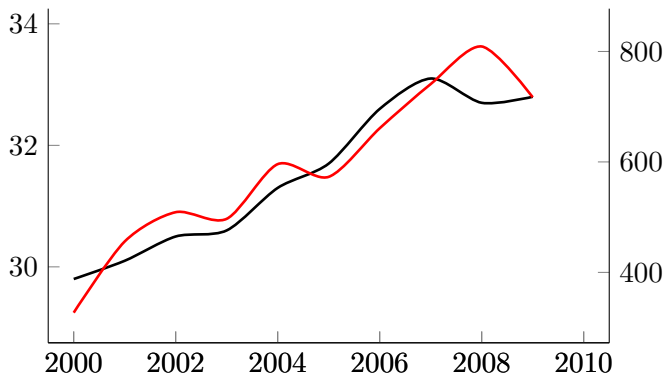
Exercise 12 - Temperature in Poland [median] - 15-minutes data



Spurious regression

- a) In case, when a model consists of non-stationary variable y (dependent variable) and if at least some x variables (independent variables) are non-stationary, distribution of test statistics are non-standard and it may lead to incorrect hypotheses testing findings.
- b) In a model with spurious regression problem t -statistic turns out to be significant, while this statistic doesn't have the Student's- t distribution, thus classical statistical testing is impossible.
- c) Estimator of β is consistent and parameters are estimated correctly, but we can not assess which are significantly different from zero with standard methods.
- d) Spurious regression problem may lead to a model in which relationships are spurious. In such cases non-stationary variables have to be transformed to stationary form.

Exercise 13 – Source: <http://www.tylervigen.com/>

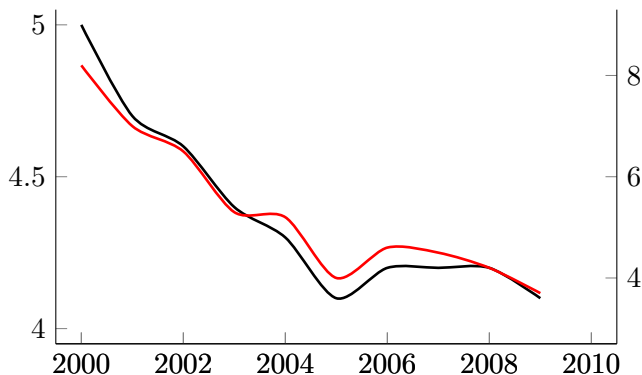


Per capita consumption of cheese (US)

Number of people who died by becoming tangled in their bedsheets

Correlation: 0.947091, $R^2 = 0.8970$

Exercise 14 – Source: <http://www.tylervigen.com/>

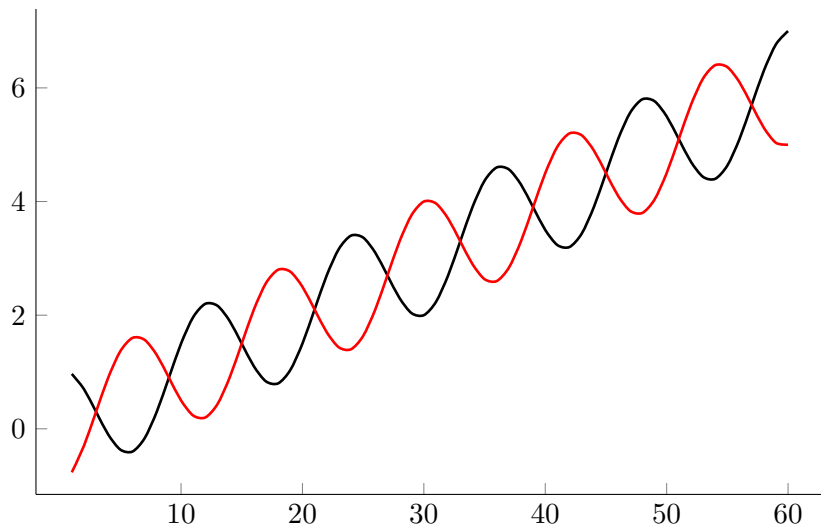


Divorce rate in Maine per 1000 people

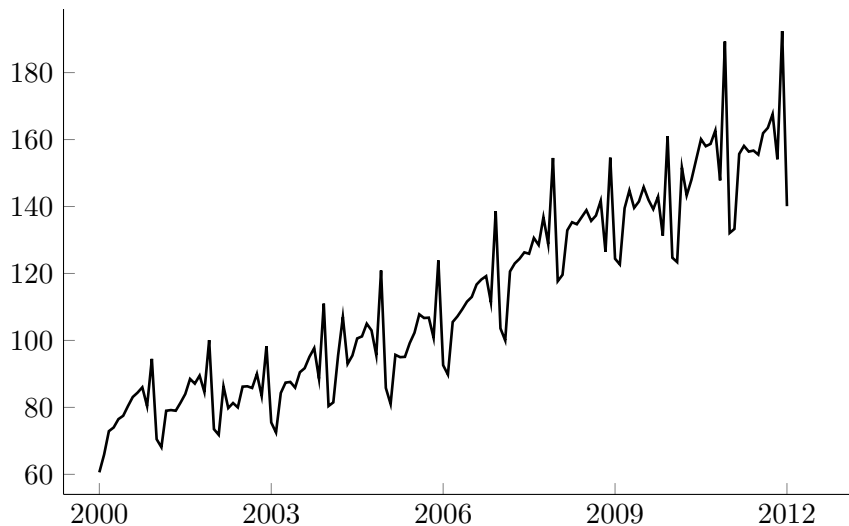
Per capita consumption of margarine

Correlation: 0.992558, $R^2 = 0.9852$

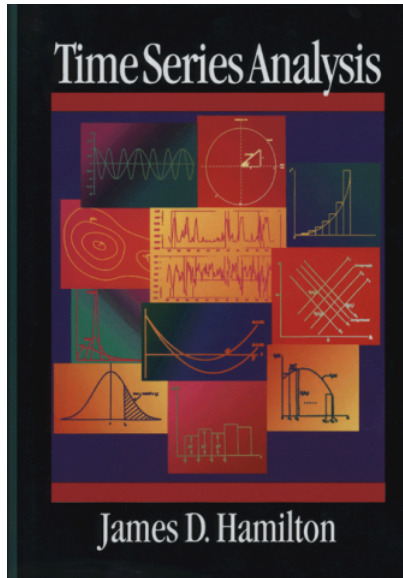
Exercise 15



Exercise 16 – Retail trade index (POL)



Time Series Analysis



Exercise 17

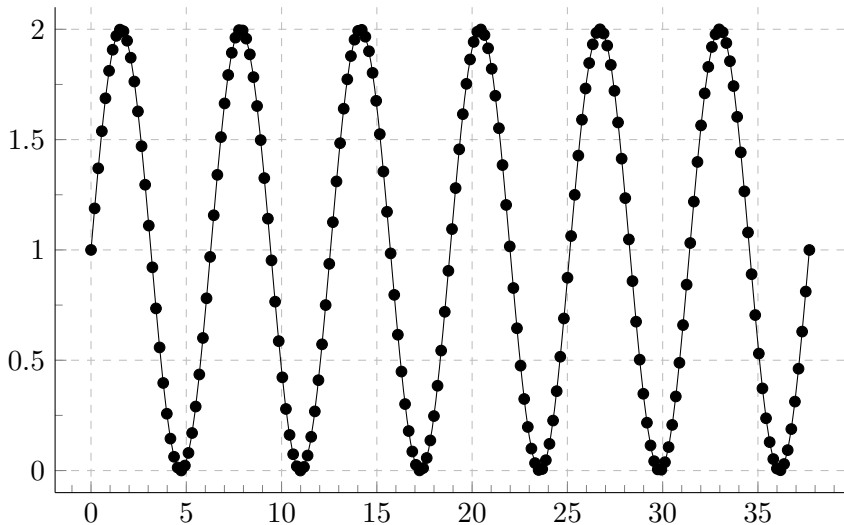
Generate $T=10,000$ realisations of the process

$$y_t = 0.99y_{t-1} + \varepsilon_t \quad (1)$$

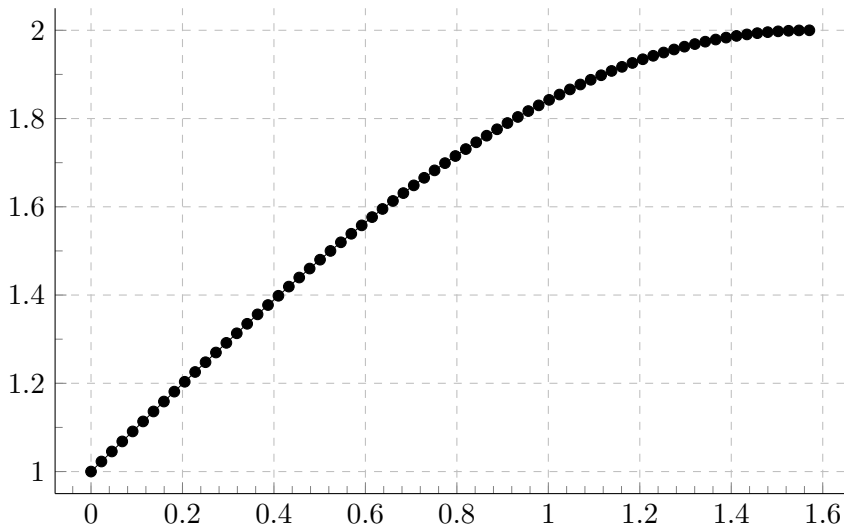
where $y_1 = 0$ and $\varepsilon_t \sim N(0, 1)$. Set seed value at 123.

- a) Plot the whole series. Is y_t stationary?
- b) Plot only first 500 observations. On the basis of the would you describe y_t as stationary?

Sinus over $[0; 12\pi]$



Sinus over $[0; \pi/2]$



The Meaning of Tests for Unit Roots

- Knowing whether nonstationarity in the data is due to a deterministic time trend or a unit root would seem to be a very important question.
- Nelson and Plosser (1982) argued that many economic series are better characterized by unit roots than by deterministic time trends.
- Nelson, Charles R., and Charles I. Plosser. 1982. "Trends and Random Walks in Macroeconomic Time Series: Some Evidence and Implications." *Journal of Monetary Economics* 10:139-62.

Source: James D. Hamilton, *Time Series Analysis*, Princeton University Press, pages 444-445.

The Meaning of Tests for Unit Roots

- Although it might be very interesting to know whether a time series has a unit root, several recent papers have argued that the question is inherently unanswerable on the basis of a finite sample observations.
- For any unit root process that will be impossible to distinguish from the unit root representation for any given sample size T . Such a stationary process is found easily enough by setting one of the eigenvalues close but not quite equal to unity.
- For example, suppose the sample consists of $T = 10,000$ observations that were really generated by a driftless random walk:

$$y_t = y_{t-1} + \varepsilon_t \quad \text{true model (unit root)}. \quad (2)$$

Consider trying to distinguish this from the following stationary process:

$$y_t = \phi y_{t-1} + \varepsilon_t \quad |\phi| < 1 \quad \text{false model (stationary)}. \quad (3)$$

Source: James D. Hamilton, *Time Series Analysis*, Princeton University Press, pages 444-445.

DF test vs. ADF test

- Dickey-Fuller Test

- $\Delta y_t = \rho y_{t-1} + \epsilon_t$
- $\Delta y_t = \alpha_1 + \rho y_{t-1} + \epsilon_t$
- $\Delta y_t = \alpha_1 + \alpha_2 t + \rho y_{t-1} + \epsilon_t$

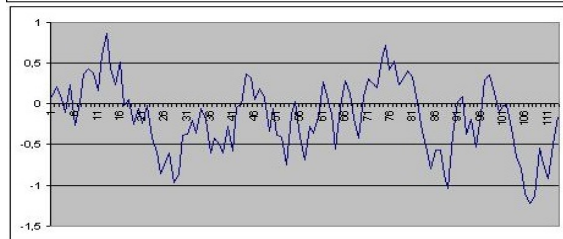
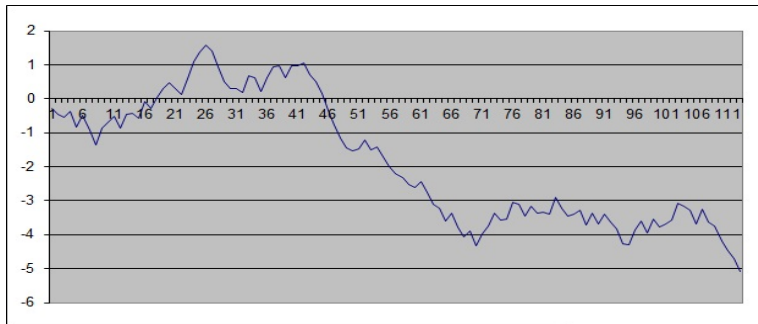
- Augmented Dickey-Fuller Test

- $\Delta y_t = \rho y_{t-1} + \sum_{i=1}^k \gamma_i \Delta y_{t-i} + \epsilon_t$
- $\Delta y_t = \alpha_1 + \rho y_{t-1} + \sum_{i=1}^k \gamma_i \Delta y_{t-i} + \epsilon_t$
- $\Delta y_t = \alpha_1 + \alpha_2 t + \rho y_{t-1} + \sum_{i=1}^k \gamma_i \Delta y_{t-i} + \epsilon_t$

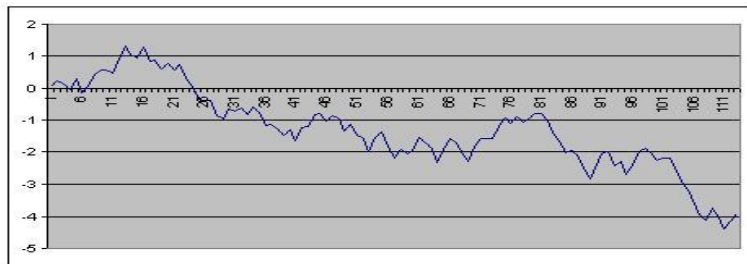
- Why do we need the ADF test? What is the reason for including lags in the equation?
- What is general purpose of lags in a model?

$$\Delta y_t = \rho y_{t-1} + \epsilon_t$$

H_0 : Random Walk, H_1 : Stationary with zero mean



$\Delta y_t = \alpha_1 + \rho y_{t-1} + \epsilon_t$, H_0 : Random Walk with drift,
 H_1 : Stationary around a constant $\neq 0$



- Kwiatkowski, Philips, Schmidt, Shin
- $y_t = \delta + \zeta_t + \epsilon_t$
- where ϵ_t is $IID(0, \sigma_\epsilon^2)$, and ζ_t is unobservable random walk:
- $\zeta_t = \zeta_{t-1} + u_t$
- $u_t \sim IID(0, \sigma_u^2)$
- $H_0 : \sigma_u^2 = 0$, variable y_t is stationary
- $H_1 : \sigma_u^2 > 0$, variable y_t is non-stationary
- **The null hypothesis is rejected when at least one test statistic is greater than the critical value.**

Time Series Analysis

Stochastic process

Stochastic process and time series

- Time series is a chronologically ordered sequence of data points, measured typically at successive times spaced at uniform time intervals.
 - These data points represent levels of some feature or characteristic measured at time intervals.
 - From the formal point of view, we assume that time series $\{y_t\}$ is a unique realization of some **stochastic (random) process**.
- Stochastic process is then defined as a series of random variables $\{Y_t\}$ indexed by time $\{t\}$
 - In practice, "time series" often refers to both its original formal meaning and stochastic process.
- Stochastic process vs. deterministic process
 - Values of realizations of deterministic processes are unambiguously defined in every moment $\{t\}$.
 - Values of realizations of non-deterministic or stochastic processes can not be unambiguously defined. Assuming a particular significance level one can only define a range covering possible values of its realizations.
- Ergodic vs. non-ergodic processes
 - **ergodic**: those, that can be described by 1 & 2 order moments,
 - **non-ergodic**: for those none of moments can be defined.



Parameters of a stochastic process

- Generally speaking, parameters of stochastic processes are functions of time:
 - Expected value of a stochastic process is a sequence of expected values $\{\mu_t\}$ of consecutive random variables $\{X_t\}$.
 - Variance of a stochastic process is a sequence of variances $\{\sigma_2^2\}$ of consecutive random variables $\{X_t\}$.
 - Covariance of a stochastic process is a variance-covariance matrix $\{\sigma_{t,t+j}\}$ describing covariances between variables $\{X_t\}$ and $\{X_{t+j}\}$

Stationarity of time series

- Strong stationarity
 - a stochastic process is strong or strict stationary if joint probability distribution does not change when shifted in time.
 - as a result, parameters such as expected value and variance, if they exist, also do not change over time.
- Formally, if:
 - $\{X_t\}$ is a stochastic process, and
 - $F_x(x_{t_1+\tau}, \dots, x_{t_k+\tau})$ represents the cumulative distribution function (CDF) of the joint distribution of $\{X_t\}$ at times $t_1 + \tau, \dots, t_k + \tau$, then $\{X_t\}$ is said to be stationary if, $\forall k, \forall \tau$:
$$F_x(x_{t_1+\tau}, \dots, x_{t_k+\tau}) = F_x(x_{t_1}, \dots, x_{t_k})$$
 - Note: since τ does not affect $F_x(\cdot)$, $F_x(\cdot)$ is not a function of time.

Weak stationarity

- In practice, much more often one considers processes that are stationary in a weak sense.
- Stochastic process is weak stationary if, for all moments t , the following conditions are fulfilled:

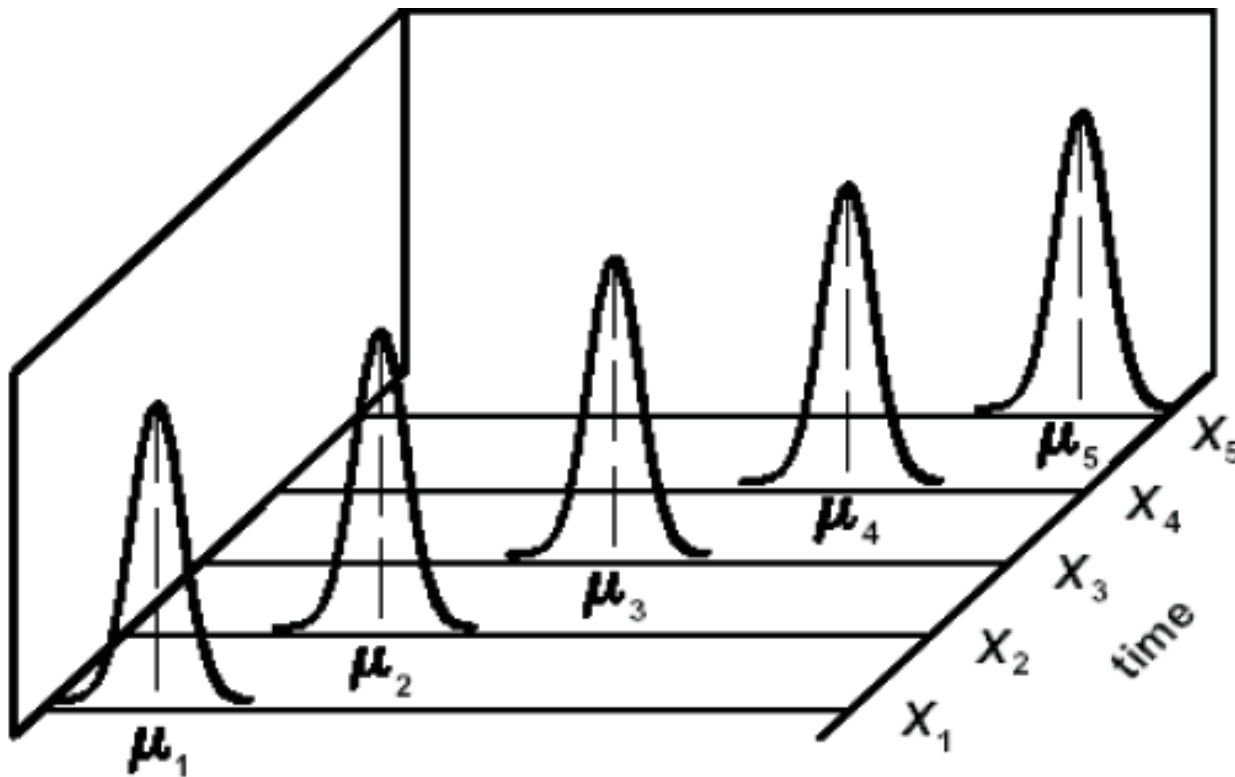
$$E(X_t) = \text{const} = \mu$$

$$\text{Var}(X_t) = \text{const} = \sigma^2$$

$$\text{Cov}(X_t, X_{t+k}) = \sigma_k^2$$

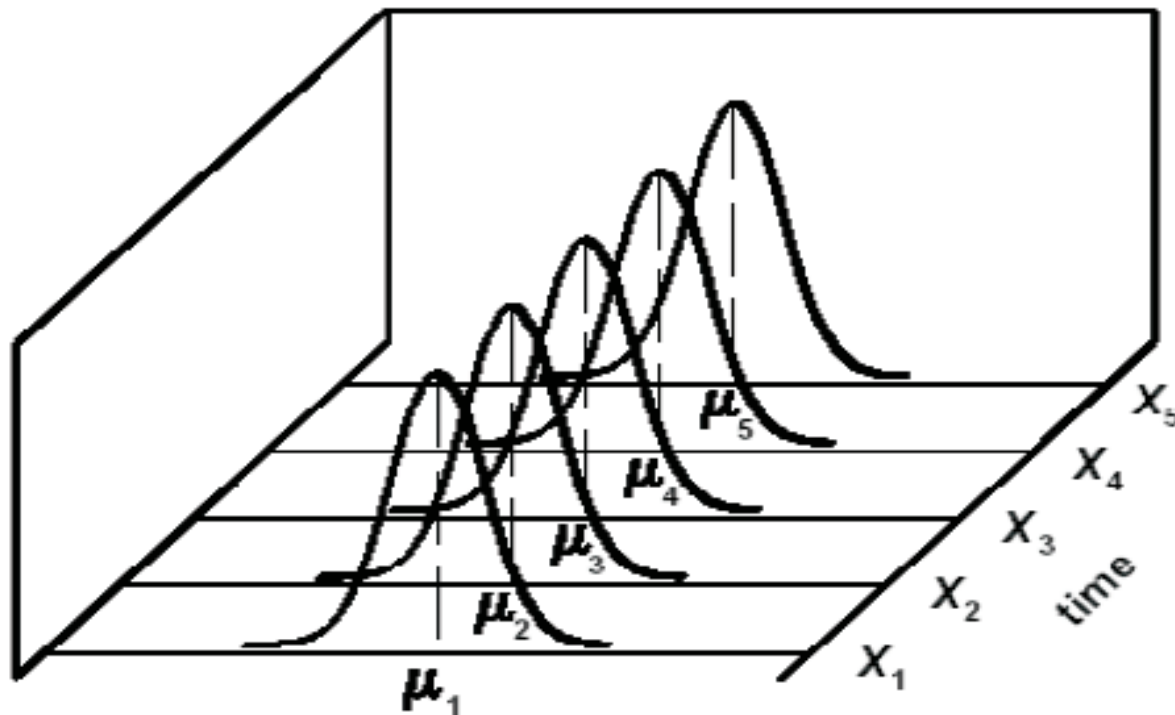
Non-stationary stochastic process

- Stochastic process with an increasing (non-stationary) expected value:



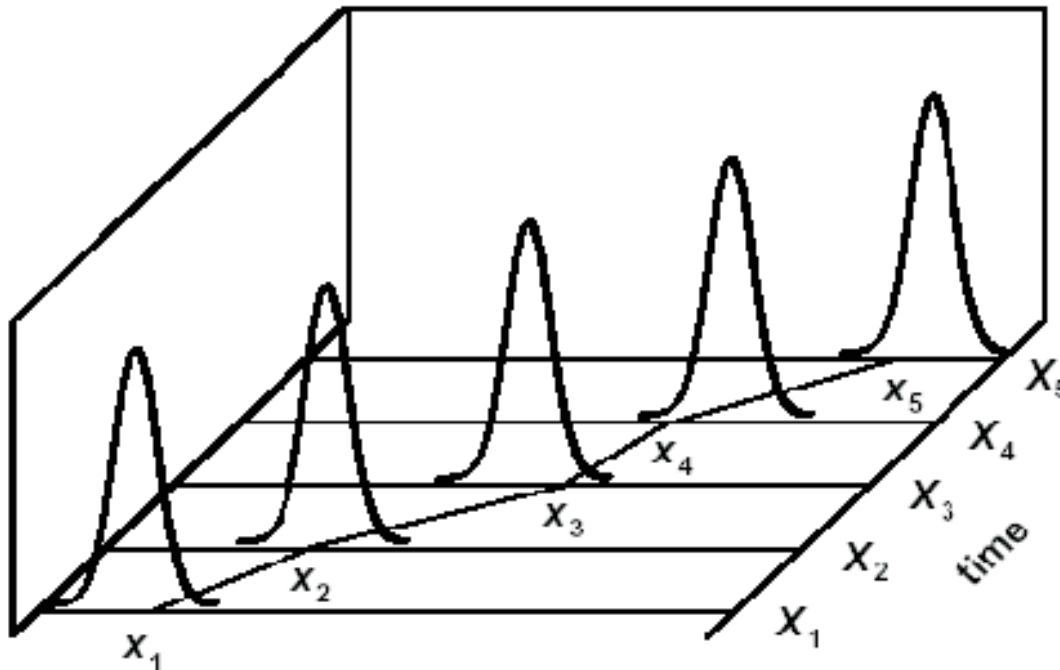
Stationary stochastic process

- Stochastic process with constant (stationary) both the expected value and the variance:



A realisation of stochastic process

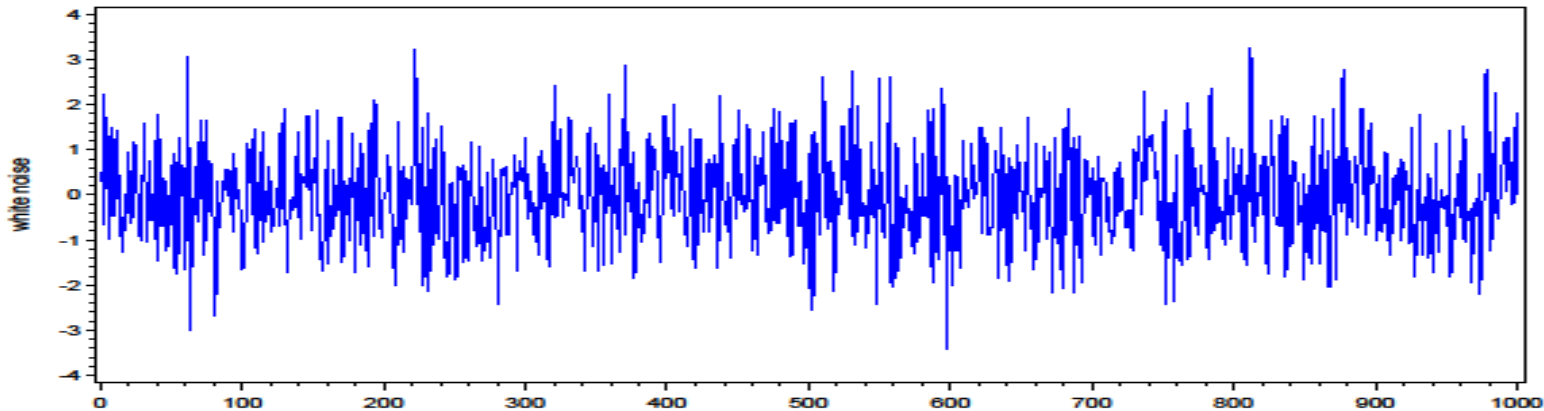
- An example of a time series -- a single realisation of a stochastic process:



- For simplicity, one often uses the term "time series" actually referring to a stochastic process.

White noise

- A special example of a stationary time series
 - It meets the following three conditions:
 - $E(X_t) = 0$,
 - $Var(X_t) = \sigma^2$,
 - $Cov(X_t, X_{t+k}) = 0$ (*exc. Var*)



White noise testing

Box-Pierce Q statistic

$$Q = T \sum_{k=1}^m \hat{\rho}_k^2 \sim \chi_m^2$$

Ljung-Box Q statistic (better for smaller samples)

$$Q^* = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k} \sim \chi_m^2$$

In both tests the null hypothesis is the same: the analyzed process is a white noise, i.e.:

$$H_0: \rho_1 = \rho_2 = \dots = \rho_m = 0$$

here ρ_k is a correlation coefficient between observation y_t and y_{t+k}

Going to the pub (Random walk) - an example of a non-stationary process

- Let's assume, that we are standing between two pubs.
- We throw a coin to decide in which direction to go:
 - heads → we make one step to the left
 - tails → we make one step to the right

$$Z_t = \begin{cases} -1 & \text{with } p = 1/2 \\ 1 & \text{with } p = 1/2 \end{cases}$$

- After each step we throw the coin again.
 - How many steps will it take us to get to our destination?
- To formally describe our location at time t we can write:

$$X_1 = Z_1$$

$$X_2 = X_1 + Z_2$$

$$X_t = \overset{\dots}{X_{t-1}} + Z_t$$

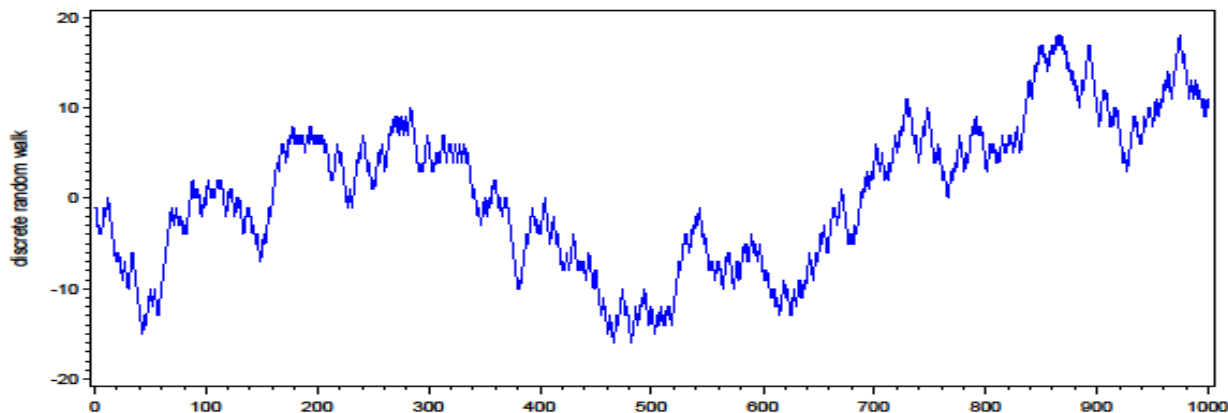
- Admittedly, we have $\mathbf{E}(X_t) = \mathbf{0}$
- But...

Going to the pub (2)

- ... variance seems to be not constant over time!

$$\begin{aligned} \text{Var}(X_1) &= \text{Var}(Z_1) = 1 \\ \text{Var}(X_2) &= \text{Var}(X_1 + Z_2) = \text{Var}(X_1) + \text{Var}(Z_2) = 2 \\ &\dots \\ \text{Var}(X_t) &= t \end{aligned}$$

- Variance of the process X_t is a linear function of time t , so X_t is a non-stationary process!
- The process X_t is an important case of a non-stationary process and is called a random walk.
 - Because the random increment is drawn from a discrete distribution, X_t is actually a discrete random walk.
- So, even if we have only 20 steps to the next pub, it could be a long way :-)



Drift in random walks

- We can also add a drift to our random walk.
- Drift means simply a systematic deviation of the same sign and size on every iteration.
- Formally, a random walk with a drift can be described as:

$$X_t = \mu + X_{t-1} + \varepsilon_t$$

- where generally $\varepsilon_t \sim IID(0, \sigma_\varepsilon^2)$.
- IID is an abbreviation from Identically and Independently Distributed.

Continuous random walks

- Continuous random walk process:

$$X_t = \mu + X_{t-1} + \varepsilon_t$$

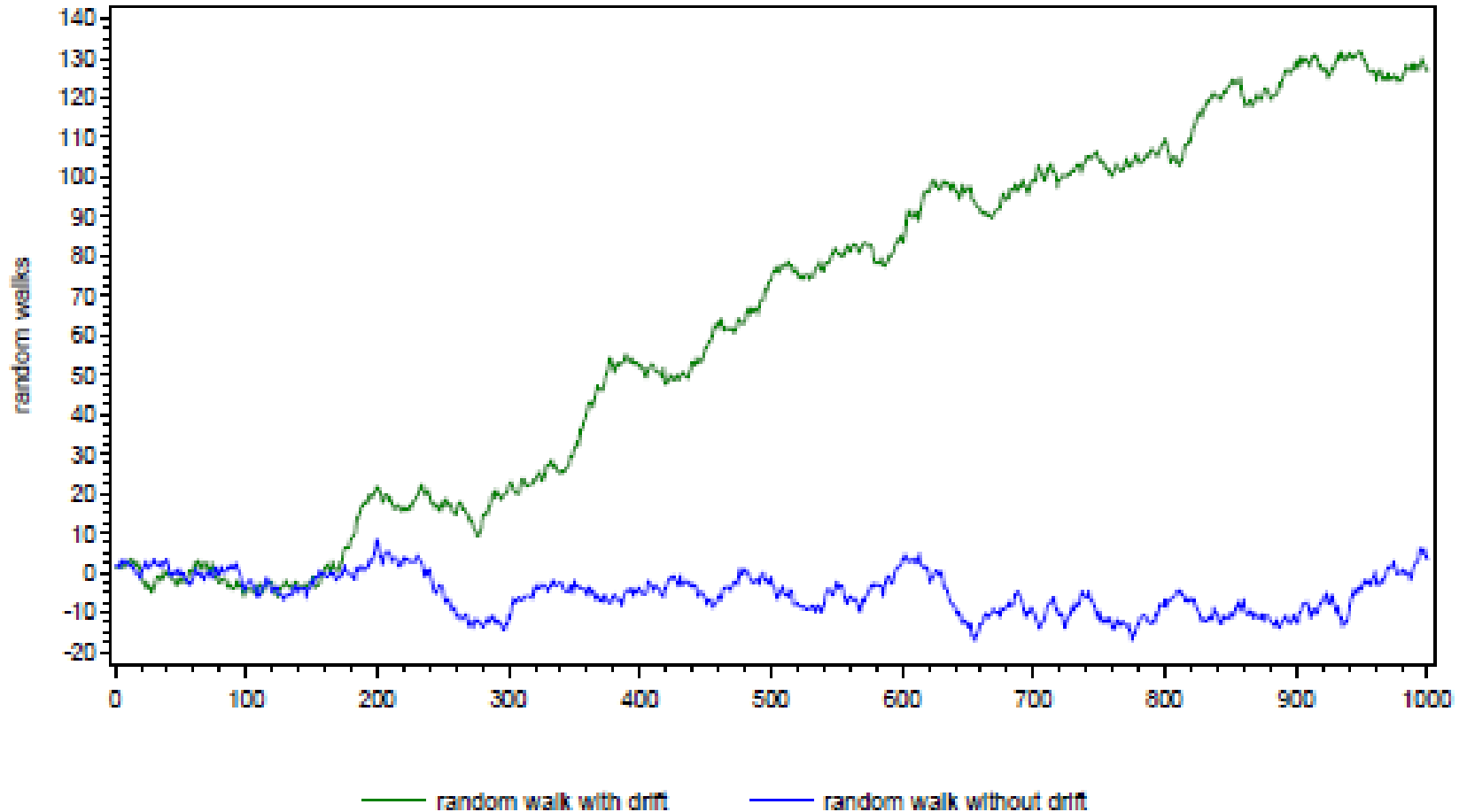
where $\varepsilon_t \sim IID(0, \sigma_\varepsilon^2)$ and additionally it is drawn from a continuous distribution.

- Continuous normal random walk process:

$$X_t = \mu + X_{t-1} + \varepsilon_t$$

where $\varepsilon_t \sim IID(0, \sigma_\varepsilon^2)$ and additionally it is drawn from a normal distribution. We can hence write $\varepsilon_t \sim NID(0, \sigma_\varepsilon^2)$, where NID stands for Normally and Independently Distributed.

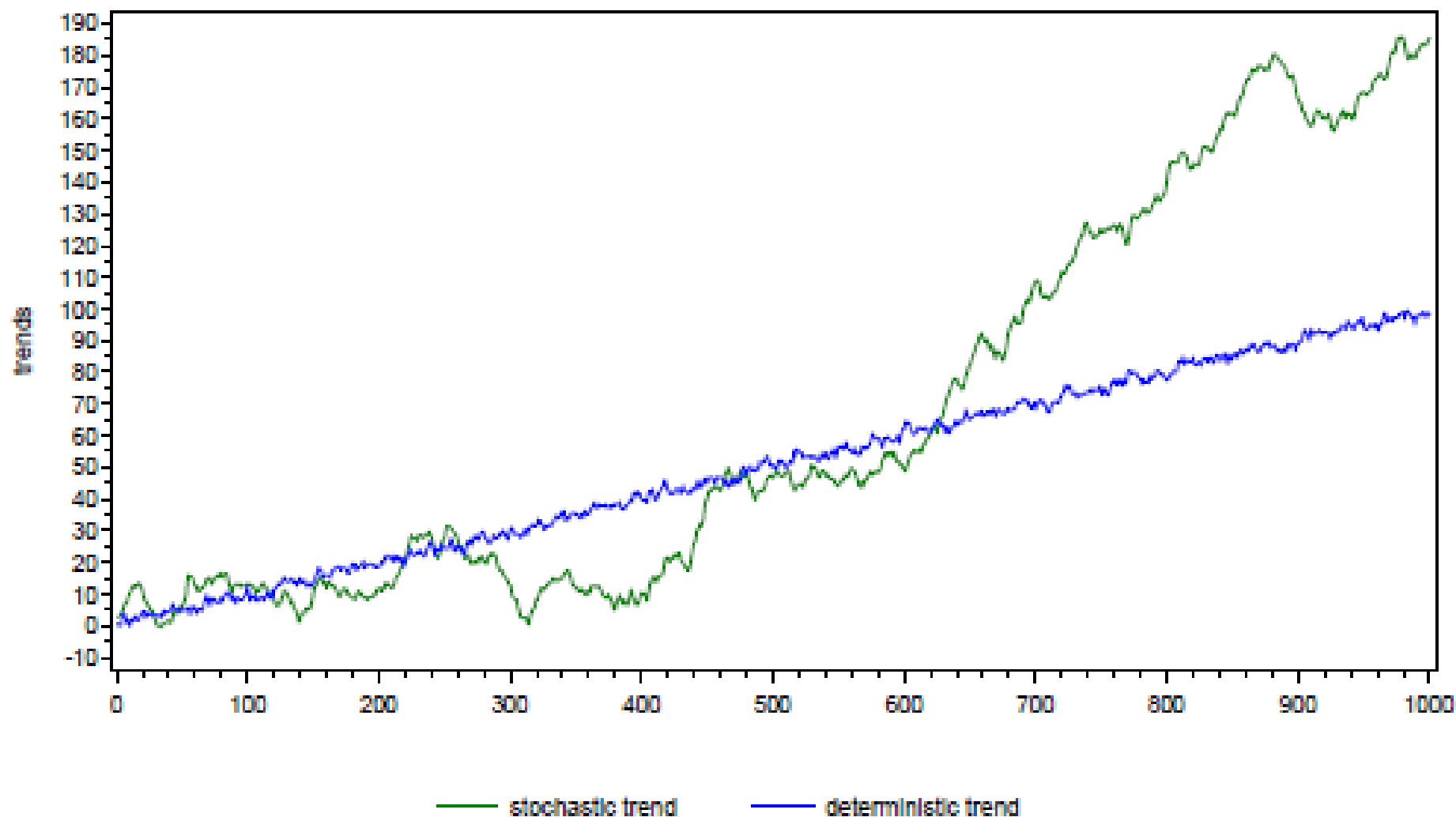
Random walks with and without a drift



Trends in time series

- Stochastic trend
 - For many economic and financial time series we can assess the type of its non-stationarity by looking on the plot of the time series.
 - If the time series has a tendency to move systematically in one direction, this means there is some kind of a trend included.
 - Time series can drift upwards or downwards only as a result of purely stochastic (random) shocks (increments). This can be referred to as time series with stochastic trends.
- Random walk
 - is an example of a time series with a stochastic trend (either with or without a drift).
 - Since variance of a random walk increases with time, during long periods values of random walk realisations can be located far away from the starting point.
- Deterministic trend
 - Another example of a tendency (trend) in a non-stationary stochastic process -- the expected value of the process might be a function of time.
 - Formally such a function can be treated as a polynomial of order k , where the argument is time. We can define it by: $S_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \dots + \beta_k t^k$ - a deterministic trend
 - Linear trend: $S_t = \beta_0 + \beta_1 t$

Stochastic vs deterministic trend



Random Walks - Conclusion

- Random walks (either with or without a drift) and deterministic trends seem to be a proper description of many economic and financial time series (there could also be a mixture of random walks and deterministic trends).
- Both kinds of stochastic processes generate non-stationary time series with strong trends.
- It is therefore not surprising that regression analysis of such time series results almost always in statistically significant results, which are treated as an evidence of strong relationships between those time series.
- However, these "relationships" (strong correlations) can emerge only as a result of similar stochastic trends, completely independently of actual link between these time series.

Spurious regressions

- Statistical properties of regression analysis based on non-stationary time series are generally doubtful.
- Additionally, there is high probability of getting statistically significant results, even if there is actually no relationship between the two variables.
- Such situations are referred to as spurious regressions,
 - For funny examples check: <http://www.tylervigen.com/spurious-correlations>
- **Regression of a quadratic trend against a linear trend**
 - $y = 1, 2, 3, \dots, n,$
 - $x = 1, 4, 9, \dots, n^2$
- Results for 50 observations:

F Value Pr > F R-Square = 0.9399 Adj R-Sq = 0.9386
750.29 <.0001

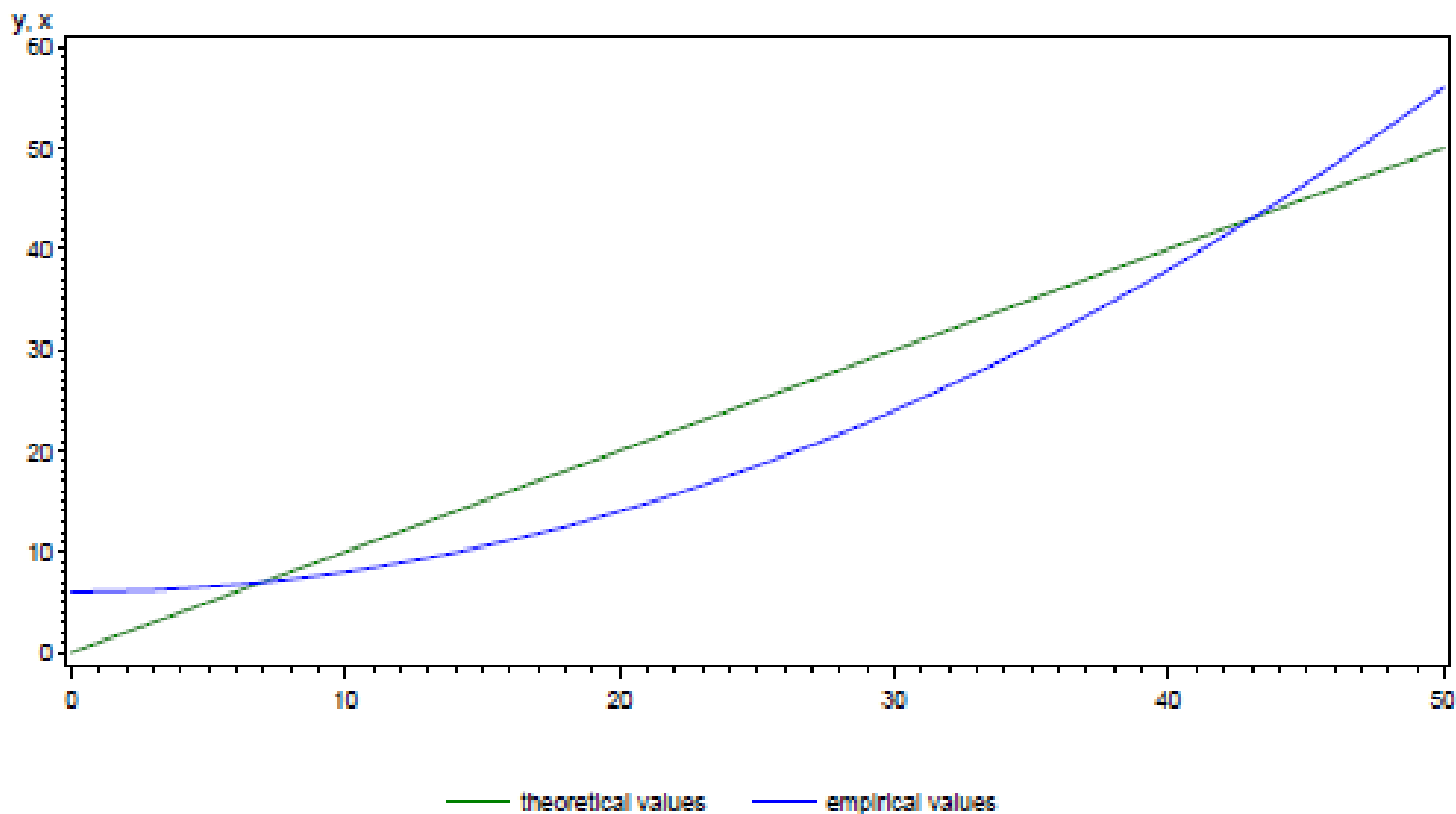
		Parameter	Standard		
Variable	DF	Estimate	Error	t Value	Pr > t
Intercept	1	9.67883	0.77103	12.55	<.0001
X	1	0.01843	0.00067	27.39	<.0001

Durbin-Watson D 0.022

- The only statistic indicating problems with that regression is a Durbin-Watson statistic!
- DW statistic close to zero means that residuals are highly autocorrelated!

Theoretical vs. actual values

- One can easily see that theoretical and actual values are not related at all.
- Only the trends are common.



Durbin – Watson test

- Durbin – Watson test is used to detect autocorrelation among residuals

$$DW = \frac{\sum_{t=2}^T (\hat{\varepsilon}_t - \hat{\varepsilon}_{t-1})^2}{\sum_{t=1}^T \hat{\varepsilon}_t^2} \in (0; 4)$$

- Value close to 2 → no autocorrelation
- Value close to 0 → positive autocorrelation
- Value close to 4 → negative autocorrelation

Spurious regressions: stochastic trends - Newbold and Davis experiment

Newbold and Davis experiment (1977):

- generate realizations of two independent non-stationary processes (random walks):

$$\begin{aligned}y_t &= y_{t-1} + \varepsilon_{1t} & \varepsilon_{1t} &\sim NID(0,1) \\x_t &= x_{t-1} + \varepsilon_{2t} & \varepsilon_{2t} &\sim NID(0,1)\end{aligned}$$

assuming that $Cov(\varepsilon_{1t}, \varepsilon_{2t}) = 0$

- next, estimate two regressions: ε_{1t} vs. ε_{2t} and y_t vs. x_t and save t and DW statistics for both regressions.
- repeat the steps a large number of times (say, 1000 times)
- as a result, having two series of t and DW statistics we calculate its mean values, standard deviations, skewness and kurtosis coefficients and we compare these values with those for t -Student distribution.
- assuming 5% level of significance we should get statistically significant results in more or less 5% of cases.
- 5% denotes here the probability of making a type I error, that is rejecting H_0 when in fact it is true (to put it in other words: the test will confirm a significant relationship, even if in reality there is no relationship at all).

Results for 1000 repetitions (ND experiment)

- Results for 1000 repetitions

	theoretical	ε_{1t} vs. ε_{2t}	y_t vs. x_t
t statistics			
mean	0.000	0.036	0.048
std. dev.	1.021	1.015	4.849
skewness	0.000	-0.151	-0.214
kurtosis	3.125	3.482	3.845
5% quantile	1.677	1.564	8.294
% of "significant"	5.000	5.2000	64.200
DW statistics			
	2.000	1.994	0.313

- x and y were generated independently, so there is no real relationship between them.
- However, results show about 64% of "significant" relationships!
- We also observe very low values for the DW statistics!

Conclusions:

- Running a regression analysis on non-stationary time series that exhibit stochastic or deterministic trends often leads to spuriously good results.
- As a result, we cannot make a proper judgement, whether relationships between variables described by the theory are also supported by empirical data.
- Hence, regression analysis should be applied only on time series that do not contain trends.
- Since many economic and financial time series are non-stationary, it is crucial to remove the trend before conducting any regression analysis.

Time Series Analysis

Stationarity testing: ADF, KPSS and PP tests

Materials are based on dr Piotr Wójcik TSA course (QF programme)

dr Marcin Chlebus, dr Rafał Woźniak

Testing stationarity in time series

- To test whether a particular time series is stationary several tests can be applied.
- The most popular are:
 - Dickey-Fuller (DF) test,
 - Augmented Dickey-Fuller (ADF) test,
 - Phillips-Perron (PP) test,
 - KPSS (Kwiatkowski-Phillips-Schmidt-Shin) test.



Notation and operators

In order to simplify the notation used, we define two kinds of operators:

- difference operator Δ

$$\Delta X_t = X_t - X_{t-1}$$

- In calculations and transformations the difference operator can be treated as a scalar.
- Here are some of its properties:

$$\Delta c = 0$$

$$\Delta^2 X = (X_t - X_{t-1}) - (X_{t-1} - X_{t-2}) = X_t - 2X_{t-1} + X_{t-2}$$

$$\Delta_2 X = X_t - X_{t-2}$$

$$\Delta X_t = (1 - L)X_t$$

- back-shift (lag) operator L

- In calculations and transformations the lag operator can be treated as a scalar.

$$L(X_t) = X_{t-1}$$

$$L^2(X_t) = L(L(X_t)) = L(X_{t-1}) = X_{t-2}$$

- relations between two operators

$$\Delta^2 X = (1 - L)^2 X_t = (1 - 2L + L^2)X_t = X_t - 2X_{t-1} + X_{t-2}$$

$$\Delta^2 X = (1 - L)^2 X_t = (1 - L)(1 - L)X_t = (1 - L)(X_t - X_{t-1}) = (X_t - X_{t-1}) - (X_{t-1} - X_{t-2})$$

Integration of time series

Definition of integration of time series by Engle and Granger (1987)

- A time series is integrated of order d if:
 - it is not stationary,
 - one can transform it into a stationary time series by differencing it d -times.
- The notation used for a time series X_t integrated of order d :

$$X_t \sim I(d)$$

- If one assumes that a and b are some parameters of a linear transformation of X_t and Y_t then one can write:
 - If $X_t \sim I(0)$ and $Y \sim I(0)$ then $(aX_t + bY_t) \sim I(0)$
 - If $X_t \sim I(0)$ and $Y \sim I(1)$ then $(aX_t + bY_t) \sim I(1)$
 - If $X_t \sim I(1)$ and $Y \sim I(1)$ then
 - $(aX_t + bY_t) \sim I(1)$ or
 - $(aX_t + bY_t) \sim I(0)$ ie. X_t and Y_t are cointegrated (details on further lectures).

Generally:

$$X_t \sim I(d) \rightarrow (a + bX_t) \sim I(d)$$

Removing trend by differencing

Differencing is a convenient way of removing trend from time series.

- If y_t is a random-walk, i.e. a non-stationary time series:

$$y_t = y_{t-1} + \varepsilon_t$$

then its first differences are stationary:

$$\Delta y_t = y_t - y_{t-1} = \varepsilon_t$$

- A time series doesn't necessarily have to be a white noise in order to be stationary!

t	y_t	Δy_t
1	$1 + \varepsilon_1$	-
2	$2 + \varepsilon_2$	$1 + (\varepsilon_2 - \varepsilon_1)$
3	$3 + \varepsilon_3$	$1 + (\varepsilon_3 - \varepsilon_2)$
4	$4 + \varepsilon_4$	$1 + (\varepsilon_4 - \varepsilon_3)$
5	$5 + \varepsilon_5$	$1 + (\varepsilon_5 - \varepsilon_4)$
6	$6 + \varepsilon_6$	$1 + (\varepsilon_6 - \varepsilon_5)$
7	$7 + \varepsilon_7$	$1 + (\varepsilon_7 - \varepsilon_6)$
...		

Differencing of order 2

In some cases calculating the first order differences is not enough to remove a trend. If so, one repeats the process of differencing and calculates the first differences of the first differences.

- If $X_t \sim I(2)$, i.e. is integrated of order 2, its first differences of the first differences will be stationary

t	y_t	Δy_t	$\Delta \Delta y_t$
1	$1 + \varepsilon_1$	-	-
2	$4 + \varepsilon_2$	$3 + (\varepsilon_2 - \varepsilon_1)$	-
3	$9 + \varepsilon_3$	$5 + (\varepsilon_3 - \varepsilon_2)$	$2 + (\varepsilon_3 - 2\varepsilon_2 + \varepsilon_1)$
4	$16 + \varepsilon_4$	$7 + (\varepsilon_4 - \varepsilon_3)$	$2 + (\varepsilon_4 - 2\varepsilon_3 + \varepsilon_2)$
5	$25 + \varepsilon_5$	$9 + (\varepsilon_5 - \varepsilon_4)$	$2 + (\varepsilon_5 - 2\varepsilon_4 + \varepsilon_3)$
6	$36 + \varepsilon_6$	$11 + (\varepsilon_6 - \varepsilon_5)$	$2 + (\varepsilon_6 - 2\varepsilon_5 + \varepsilon_4)$
7	$49 + \varepsilon_7$	$13 + (\varepsilon_7 - \varepsilon_6)$	$2 + (\varepsilon_7 - 2\varepsilon_6 + \varepsilon_5)$
...			

Dickey-Fuller test

One of the most popular tests used to determine the order of integration of a time series is the Dickey-Fuller (DF) test

- The main idea is to test whether a particular time series y_t is described by a random walk process:

$$y_t = y_{t-1} + \varepsilon_t$$

- The problem is, that we cannot directly test the hypothesis $H0: \rho = 1$ in the following equation:

$$y_t = \rho y_{t-1} + \varepsilon_t$$

because if $H0$ is true, we will be dealing with non-stationary time series and there will be high probability of obtaining spurious results.

- Dickey and Fuller proposed to transform the equation into:

$$y_t = (1 - \delta)y_{t-1} + \varepsilon_t$$

which can also be written as:

$$\Delta y_t = \delta y_{t-1} + \varepsilon_t$$

- The tested hypotheses will be simply:

$$H0: \delta = 0$$

$$H1: \delta \neq 0$$

Dickey-Fuller test

If the null hypothesis is not rejected $H0: \delta = 0$ the variable Δy_t is stationary and the testing equation is reduced to:

$$\Delta y_t = \varepsilon_t$$

- The test statistic is defined as:

$$DF = \frac{\delta_{OLS}}{S(\delta_{OLS})}$$

- The tested null hypothesis:

$$H0: y_t \text{ is at least } \sim I(1)$$

$$H1: y_t \text{ is } \sim I(0)$$

Note: DF statistics do not have a t-Student distribution. Their critical values usually are determined using Monte-Carlo simulations

Testing integration of higher orders

Let's assume that $y_t \sim I(2)$, i.e.

$$\Delta^2 y_t = \varepsilon_t$$

In such a situation we first test:

$$H0: y_t \sim I(1)$$

$$H1: y_t \sim I(0)$$

- with the testing equation:

$$\Delta y_t = \delta y_{t-1} + \varepsilon_t$$

If we cannot reject $H0$ we difference y_t and test:

$$H0: y_t \sim I(2) \rightarrow \Delta y_t \sim I(1)$$

$$H1: y_t \sim I(2) \rightarrow \Delta y_t \sim I(0)$$

- with the testing equation:

$$\Delta \Delta y_t = \delta \Delta y_{t-1} + \varepsilon_t$$

Distribution of the DF statistic

We define a *Data Generating Process*:

$$y_t = y_{t-1} + u_t, u_t \sim NID(0,1)$$

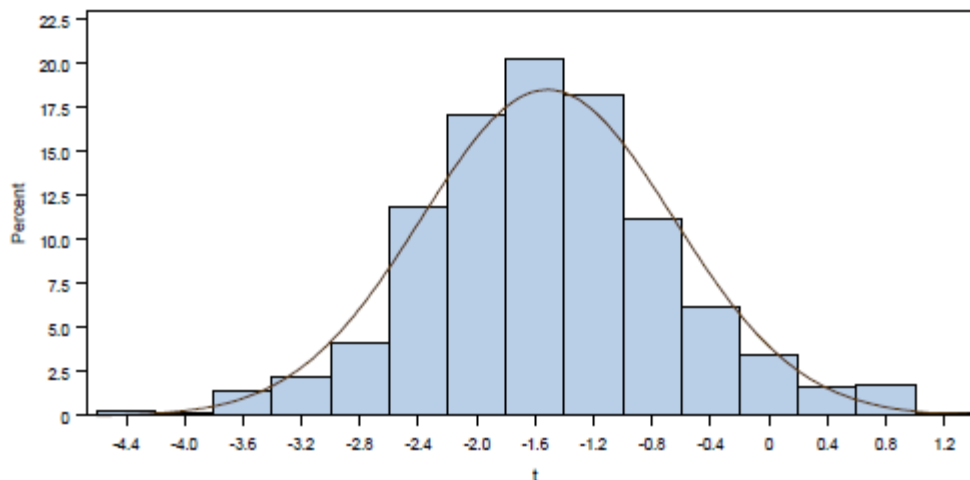
- For a selected number of observations T we run a large number (m) of simulations of the process y_t .
- For each simulated set of values of y_t we estimate the model:

$$\Delta y_t = \delta y_{t-1} + \varepsilon_t$$

- Finally, we calculate and save m t-Student statistics. Using empirical distribution of t statistics we are testing whether ρ is negative (lay in 5% extreme negative observations o t statistics distribution).

Critical Value for DF statistics calculation

T=100, m=1000	N(0,1)	t(99)	DF
mean	0	0	-1,519
st. dev.	1	1	0,862
skewness	0	0	0,196
kurtosis	3	3,063	3,577
1% quantile	-2,326	-2,365	-3,52
5% quantile	-1,645	-1,66	-2,903
10% quantile	-1,281	-1,29	-2,543



Limitations of the DF test

- Important limitation of using the DF test: residuals ε_t in testing equation must be a white noise process! They cannot be autocorrelated!
- If residuals are autocorrelated an estimator of the parameter δ is not efficient and its standard error is actually greater than we assume.

Augmented DF test

The easiest way to remove autocorrelation from residuals is to include one or more lagged values of the dependent variable among regressors in the DF testing equation:

$$\Delta y_t = \delta y_{t-1} + \sum_{i=1}^k \delta_i \Delta y_{t-i} + \varepsilon_t$$

- Lagged values of the dependent variable in the DF test are referred to as augmentations.

How to determine a proper number of augmentations k ?

- If k is too small, critical values of the DF will be improper -- we will reject H_0 too often.
- If K is too large, the power of the DF test will be low -- we will reject H_0 too rarely (power of the test is the probability of rejecting a false null hypothesis).

A common practice:

- We start with no augmentations at all and check whether residuals in the DF testing equation are autocorrelated.
- If they are, we add one augmentation and check whether it removed autocorrelation from residuals.
- If they are still autocorrelated, we add another augmentation and repeat these steps until residuals are white noise.

An example of ADF test (1)

Below one can find the output from the testdf function:

- To test whether residuals in the testing equation are white noise, a Breusch-Godfrey (BG) statistic is used.
- In the following example, residuals are autocorrelated until 2 augmentations are added.
- Augmenting the testing equation with lagged (of order 2) values of the dependent variable removes autocorrelation.
- Hence, the proper value of the ADF statistic to test the integration order is 0.27418797 which does not reject H_0 (p-value=0.976401).

	augmentations	adf	p_adf	bgodfrey.statistic	bgodfrey.statistic.1	bgodfrey.statistic.2	p_bg.p.value.1	p_bg.p.value.2
1	0	-0.02327215	0.9537731	4324	17.140403811619	25.994309992	0.00003471589	0.000002266769
2	1	0.15417877	0.9681014	4323	0.037285075608	9.080113222	0.84688579206	0.010672802337
3	2	0.27418797	0.9764016	4322	0.000353396812	0.005701143	0.98500157014	0.997153487580
4	3	0.29058538	0.9770663	4321	0.000008763535	0.002948205	0.99763800439	0.998526983558

An example of ADF test (2)

Since we did not reject H_0 in the next stage we calculate first differences of the dependent variable and repeat the whole procedure.

- After differencing the dependent variable, we obtain following results
- This time only one augmentation should be added the testing equation
- Hence, the proper value of the ADF statistic here is -50.17631. Its p-value is lower than 0.01 so we can easily reject H_0 .
- Final conclusion: since first differences of the dependent variable are stationary, i.e. $I(0)$, the dependent variable itself is integrated of order 1, i.e. $I(1)$.

	augmentations	adf	p_adf	bgodfrey.statistic	bgodfrey.statistic.1	bgodfrey.statistic.2	p_bg.p.value.1	p_bg.p.value.2
1	0	-70.12491	0.01	4323	0.037016547230	9.038613083	0.8474314	0.01089658
2	1	-50.17631	0.01	4322	0.000332295348	0.005470827	0.9854562	0.99726832
3	2	-40.11644	0.01	4321	0.000007526182	0.002827578	0.9978111	0.99858721
4	3	-34.70090	0.01	4320	0.005821367050	0.006136224	0.9391821	0.99693659

Two kinds of trends

Generally, one can define two types of nonstationarity

- random walk with or without drift (difference-stationary process)

$$y_t = \mu + y_{t-1} + \varepsilon_t$$

- deterministic trend (deterministic non-stationarity, trend-stationary process)

$$y_t = \alpha + \beta t + \varepsilon_t$$

- Although difference-stationary and trend-stationary series are both "trending" over time, the two will require different treatments to induce stationarity.

If it is believed that the series is trend-stationary, a regression of the form given in the 2 equation should be run, and any subsequent estimation should be done on the residuals from it, which would have had the linear trend removed.

If a particular time series is believed to contain only stochastic trends, then the solution is to difference it once or twice (if needed).

Two kinds of trends

On the other hand, if the first differences of a trend-stationary series were calculated, it would "remove" the non-stationarity, but at the expense of introducing an MA(1) structure into the errors.

$$y_t = \alpha + \beta t + \varepsilon_t$$

This model can be expressed for time $t-1$:

$$y_{t-1} = \alpha + \beta(t-1) + \varepsilon_{t-1}$$

Subtracting $t-1$ from t gives:

$$\Delta y_t = \beta + (\varepsilon_t - \varepsilon_{t-1})$$

- A moving average in the errors that has been created is a non-invertible MA (i.e. one that cannot be expressed as an autoregressive process). Thus, in general, the series y_t would in this case have some very undesirable properties.
- Conversely, if one would try to de-trend a series with a stochastic (rather than deterministic) trend, the non-stationarity would not be removed.
 - One possibility is to **nest both cases in a more general model**.
- This is behind the idea of three versions of the testing equation in the DF/ADF test.

3 versions of ADF test

Generally, the (A)DF test has three versions:

$$\Delta y_t = \delta y_{t-1} + \varepsilon_t$$

$$\Delta y_t = \mu + \delta y_{t-1} + \varepsilon_t$$

$$\Delta y_t = \mu + \beta t + \delta y_{t-1} + \varepsilon_t$$

- This means that one can additionally assume that y_t has a drift and/or a deterministic trend.
- So, finally, one can write the testing equation in the ADF test as:

$$\Delta y_t = \mu + \beta t + \delta y_{t-1} + \sum_{i=1}^k \delta_i \Delta y_{t-i} + \varepsilon_t$$

- where, as earlier, the tested hypothesis is:

$$H0: y_t \text{ is at least } \sim I(1)$$

$$H1: y_t \text{ is } \sim I(0)$$

Disadvantages of the ADF test

- Low power when a sample is small.
- Low power when a time series is stationary but the ρ parameter is close to 1, e.g.:

$$y_t = \mu + 0,97y_{t-1} + \varepsilon_t$$

Low power means that the test will reject H_0 too rarely, when it is actually not true.

Phillips-Perron test

Phillips and Perron (1988) developed a number of unit root tests:

- the PP tests differ from the ADF tests mainly in how they deal with serial correlation and heteroskedasticity in the errors of a testing equation,
- the PP tests correct for any serial correlation and heteroskedasticity in the errors of the test regression by directly modifying the test statistics,
- Under the null hypothesis on non-stationarity the PP test statistics have the **same asymptotic distributions as the ADF t-statistic**,
- one advantage of the PP tests over the ADF tests is that the PP tests are robust to general forms of heteroskedasticity in the error term,
- Another advantage is that the user **does not have to specify a lag length** for the test regression.

Kwiatkowski, Phillips, Schmidt and Shin (KPSS) test

Another solution to low power of the ADF test is reverting a tested hypothesis.

- An example is the KPSS test, which tests the following hypothesis:

$$H_0: y_t \text{ is } \sim I(0)$$

$$H_1: y_t \text{ is at least } \sim I(1)$$

- The KPSS stationarity test is a one-sided right-tailed test so that one rejects the null of stationarity at the α level if the KPSS test statistic is **greater than the appropriate critical value**

Problems with unit root tests

- The ADF and PP tests are asymptotically equivalent but may differ substantially in finite samples due to the different ways in which they correct for serial correlation in the test regression.
- In general, the ADF and PP tests have very low power against $I(0)$ alternatives that are close to being $I(1)$.
- Also, the power of unit root tests diminish as deterministic terms are added to the test regressions -- tests that include a constant and trend in the test regression have less power than tests that only include a constant in the test regression.